CurricuForge Documentation

CurricuForge is a state-of-the-art, AI-powered curriculum generation platform designed for modern education. It empowers faculty to instantly generate structured, high-quality learning paths and allows students to access expert-crafted content in a sleek, interactive environment.

Scenario1:AI Curriculum Generation by Faculty

In this scenario, a faculty member logs into the system and navigates to the AI Curriculum Generator section. The faculty enters details such as the subject, course duration, and learning level. Once the information is submitted, the system sends the request to the AI model integrated through Hugging Face. The AI processes the input and generates a structured curriculum that includes the course overview, learning objectives, and a week-by-week breakdown of topics, activities, and assessments in a detailed table format. The faculty member can review the generated curriculum, make any necessary adjustments, and publish it so that students can access the learning path on their dashboard.

Scenario 2: Manual Course Creation and Publishing by Faculty

In this scenario, a faculty member manually creates a course using video resources. After logging into the platform, the faculty navigates to the Content Management section and selects the option to add a new video course. The faculty provides the course title, description, YouTube video link, and additional details using markdown content for better formatting. Initially, the course is saved as a draft so the faculty can review the information. Once the content is finalized, the faculty clicks the publish option. The course then becomes visible to students on the platform, allowing them to access the video-based learning materials.

Scenario 3: Student Access and Course Discovery

In this scenario, a student logs into the platform and accesses the student dashboard. The dashboard allows the student to explore learning content through two sections: AI Learning Paths and Video Courses. Using the advanced search and filtering options, the student can find courses based on subject, learning level, or specific keywords. After selecting a course, the student can view detailed curriculum information, including weekly topics, learning activities, and assessments or video content. This structured interface helps the student easily discover and follow courses that match their learning needs.

Key Features

1. AI Curriculum Generation

Faculty can generate comprehensive curricula by simply entering a subject, duration, and learning level. Our AI (integrated via Hugging Face) produces:

- Course Overview & Objectives
- Week-by-Week Breakdown
- Detailed Tables with Topics, Activities, and Assessments

2. Content Management

- Manual Course Addition: Faculty can manually add video-based courses (YouTube) with detailed descriptions and markdown content.
- Publishing Workflow: One-click publishing to make content visible to students.
- Unified History: A tabbed history view to manage both AI-generated and manual courses separately.

3. Student Dashboard

- Tabbed Discovery: Switch between "AI Learning Paths" and "Video Courses".
- Advanced Search & Filtering: Find content by subject, level, or keywords.
- Detailed View: Rich, responsive viewing experience for all curriculum content.

4. Secure User System

- Role-Based Access: Specialized interfaces for Faculty and Students.
- Smart Authentication: Glassmorphic login and registration flows.
- Password Recovery: Secure 6-digit OTP system with email verification for password resets.

Technology Stack

Component	Technology
Frontend	React, Vite, Lucide Icons, Vanilla CSS
Backend	Flask (Python), MongoDB (PyMongo)
Authentication	JWT (JSON Web Tokens), Bcrypt Hashing
AI Engine	Hugging Face Inference API
Email Service	SMTP (Gmail App Passwords)

Setup Instructions

Backend Setup

1. Clone the repository and navigate to the backend folder.
2. Create a virtual environment: python -m venv venv.
3. Activate it: venv\Scripts\activate (Windows) or source venv/bin/activate (Mac/Linux).
4. Install dependencies: pip install flask flask-cors pymongo flask-jwt-extended bcrypt python-dotenv requests.
5. Create a .env file with:

- MONGO_URI
- JWT_SECRET_KEY
- HUGGINGFACE_API_KEY
- MAIL_SERVER, MAIL_PORT, MAIL_USERNAME, MAIL_PASSWORD (for OTPs).

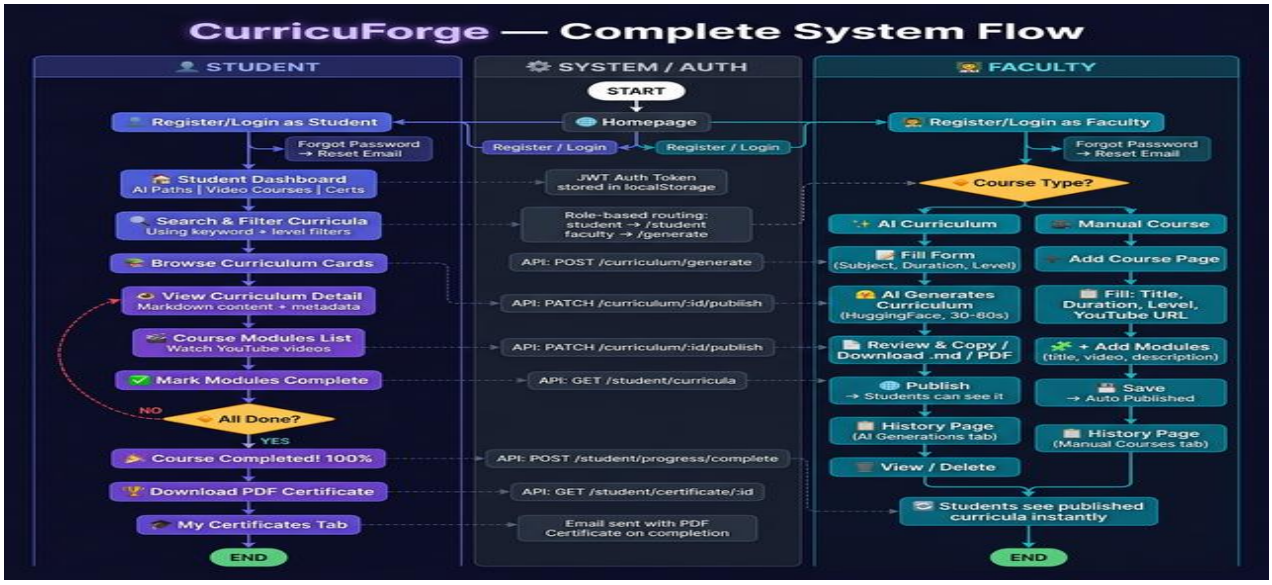
6. Run the server: python app.py.

Frontend Setup

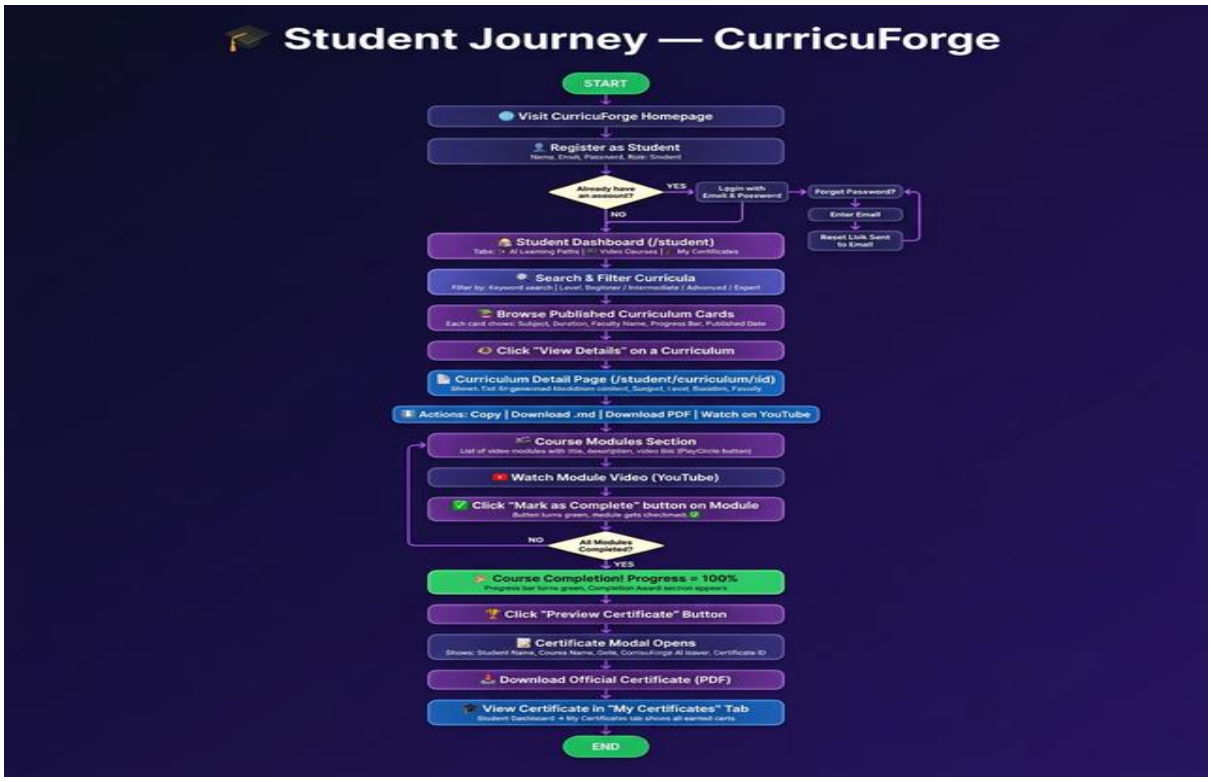
1. Navigate to the frontend folder.
2. Install dependencies: npm install.
3. Start the development server: npm run dev.
4. Access the app at http://localhost:5173.

Flow Charts

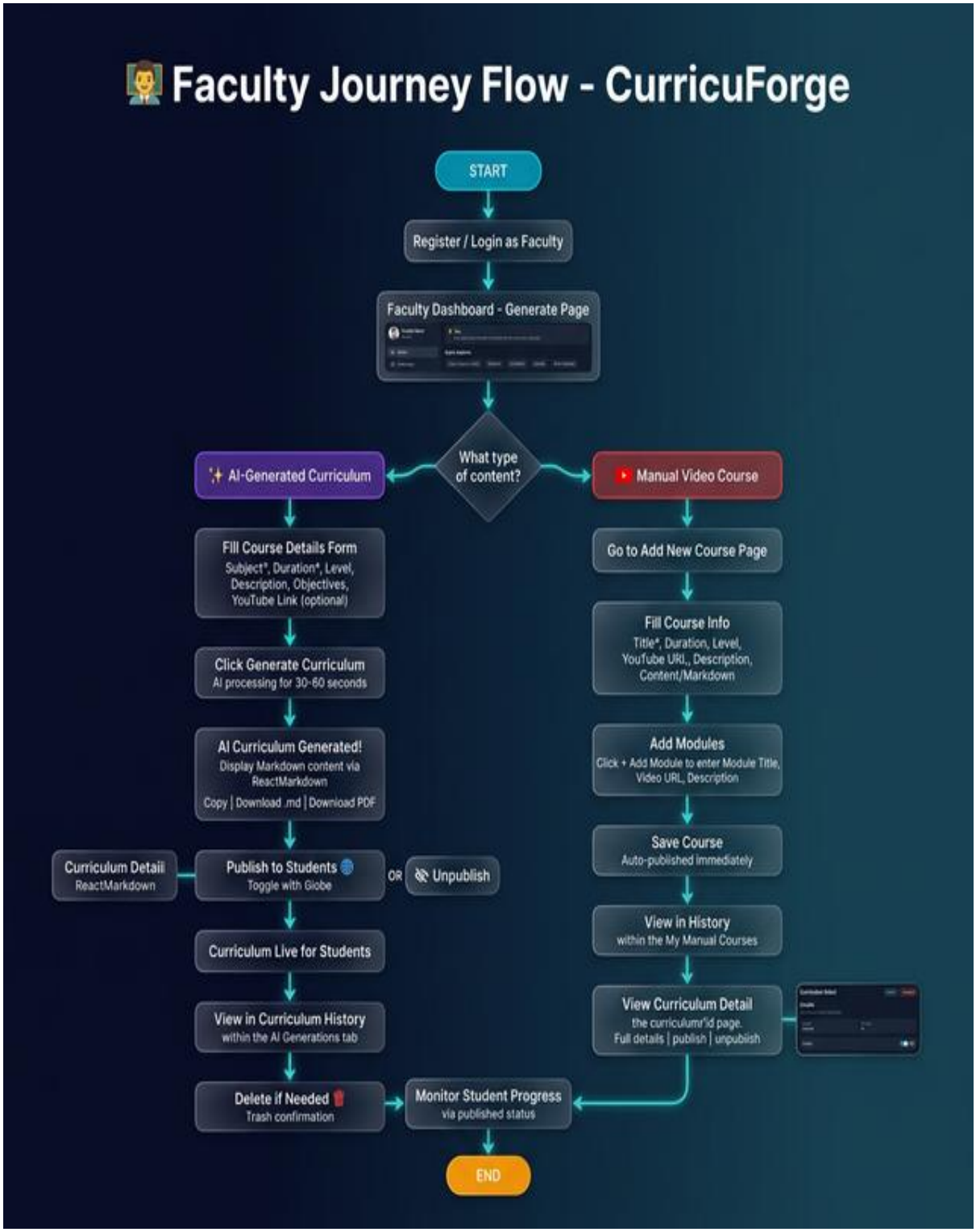
Project Flow chart



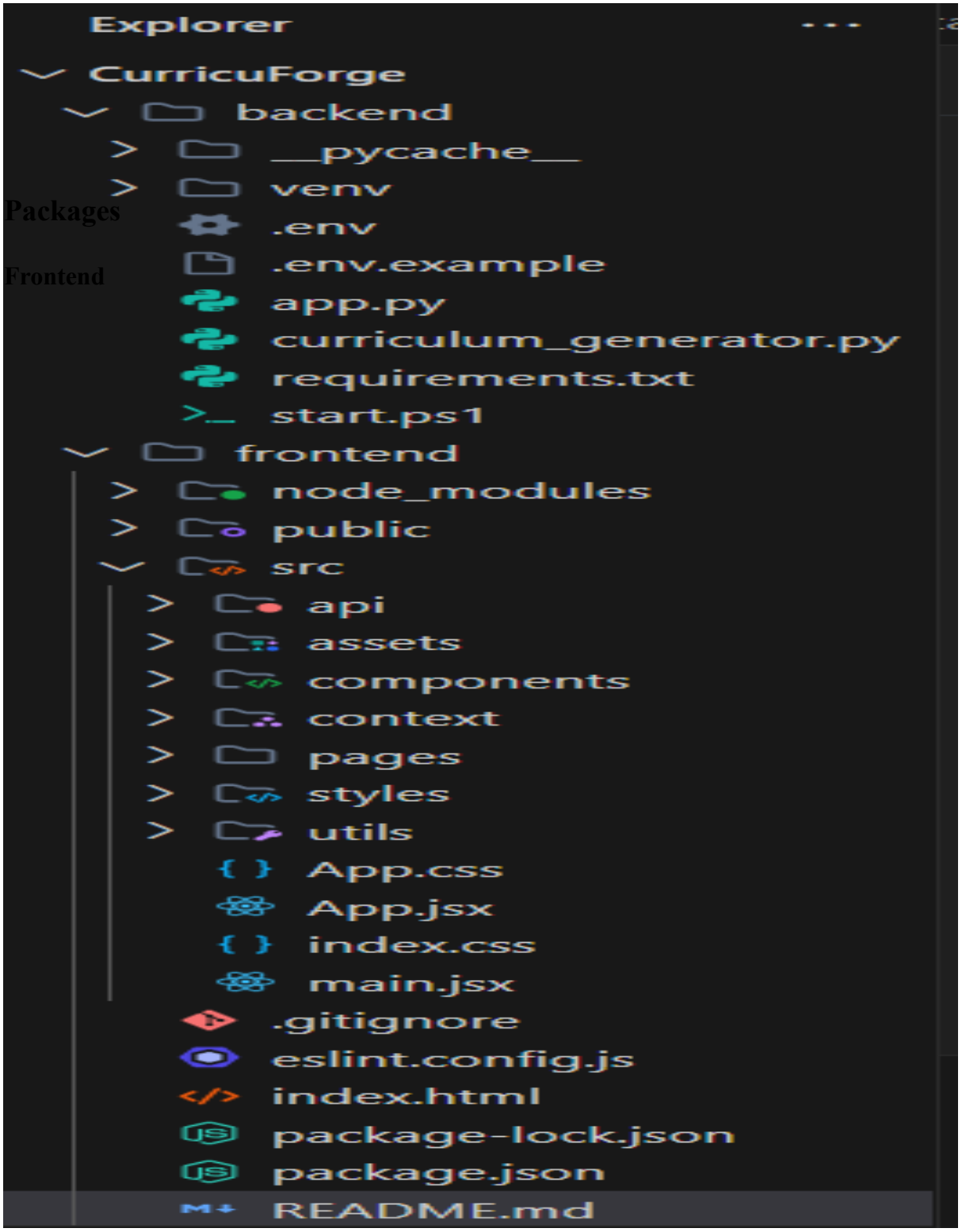
Student



Faculty


Faculty Journey Flow - CurricuForge
START
Register / Login as Faculty
Faculty Dashboard - Generate Page
What type of content?
AI-Generated Curriculum
Manual Video Course
Fill Course Details Form
Subject*, Duration*, Level, Description, Objectives, YouTube Link (optional)
Click Generate Curriculum
AI processing for 30-60 seconds
AI Curriculum Generated!
Display Markdown content via ReactMarkdown
Copy | Download .md | Download PDF
Curriculum Detail
ReactMarkdown
Publish to Students
Toggle with Globe
OR
Unpublish
Curriculum Live for Students
View in Curriculum History
within the AI Generations tab
Delete if Needed
Trash confirmation
Go to Add New Course Page
Fill Course Info
Title*, Duration, Level, YouTube URL, Description, Content/Markdown
Add Modules
Click + Add Module to enter Module Title, Video URL, Description
Save Course
Auto-published immediately
View in History
within the My Manual Courses
View Curriculum Detail
the curriculum*id page.
Full details | publish | unpublish
Monitor Student Progress
via published status
END

Project Structure



Packages

Frontend

Frontend

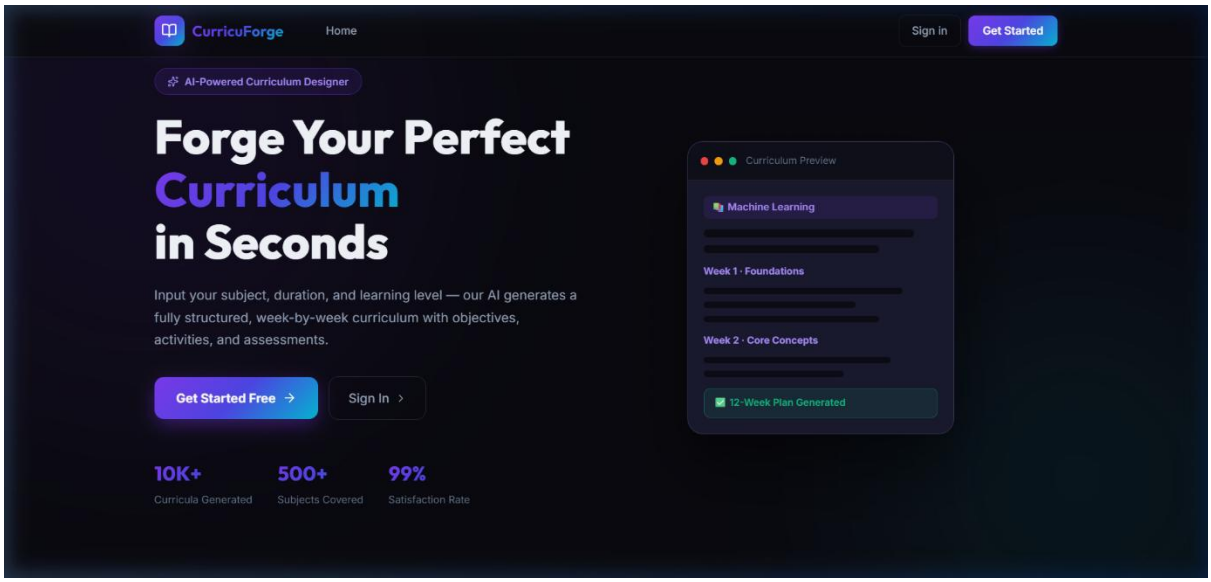
```
{
  "name": "frontend",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  ▷Debug
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "axios": "^1.13.6",
    "html2canvas": "^1.4.1",
    "jspdf": "^4.2.0",
    "lucide-react": "^0.577.0",
    "react": "^18.3.1",
    "react-dom": "^18.3.1",
    "react-markdown": "^10.1.0",
    "react-router-dom": "^7.13.1",
    "remark-gfm": "^4.0.1"
  },
  "devDependencies": {
    "@eslint/js": "^9.13.0",
    "@types/react": "^18.3.12",
    "@types/react-dom": "^18.3.1",
    "@vitejs/plugin-react": "^4.3.3",
    "eslint": "^9.13.0",
    "eslint-plugin-react": "^7.37.2",
    "eslint-plugin-react-hooks": "^5.0.0",
    "eslint-plugin-react-refresh": "^0.4.14",
    "globals": "^15.11.0",
    "vite": "^5.4.10"
  }
}
```

Backend

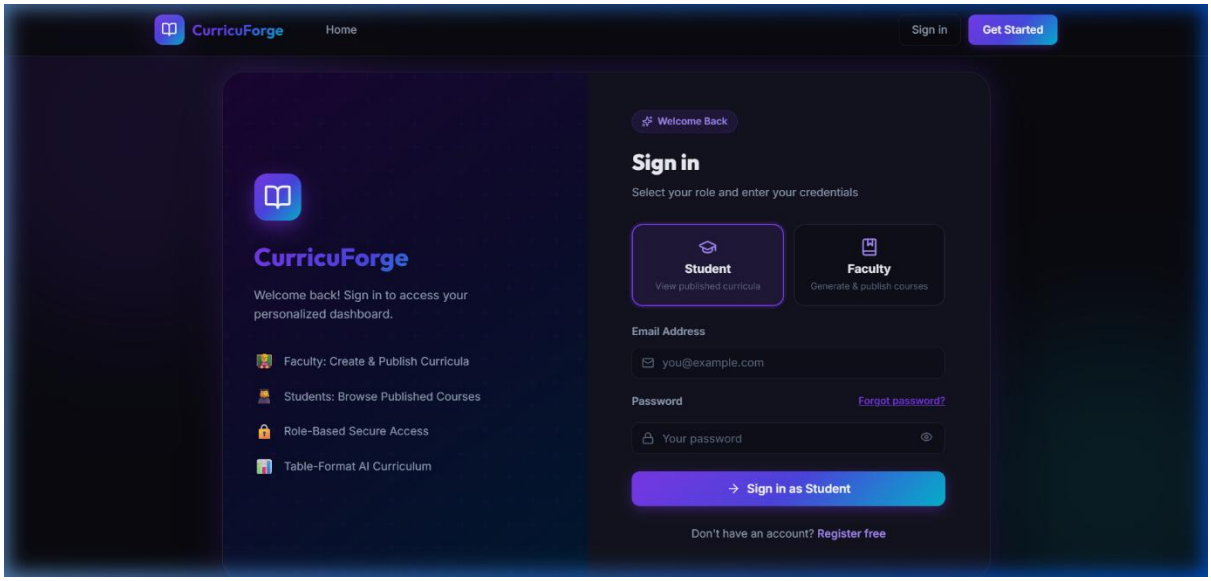
```
flask==3.0.0
flask-cors==4.0.0
flask-jwt-extended==4.6.0
pymongo==4.6.1
python-dotenv==1.0.0
bcrypt==4.1.2
requests==2.31.0
dnspython==2.4.2
```

Project Showcase

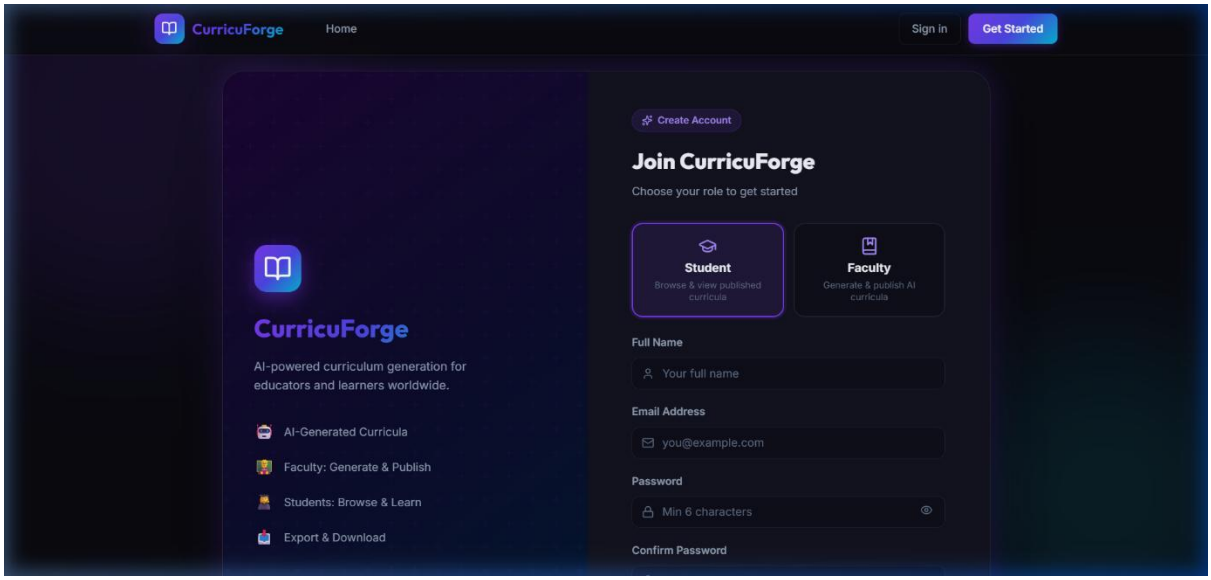
Landing Page



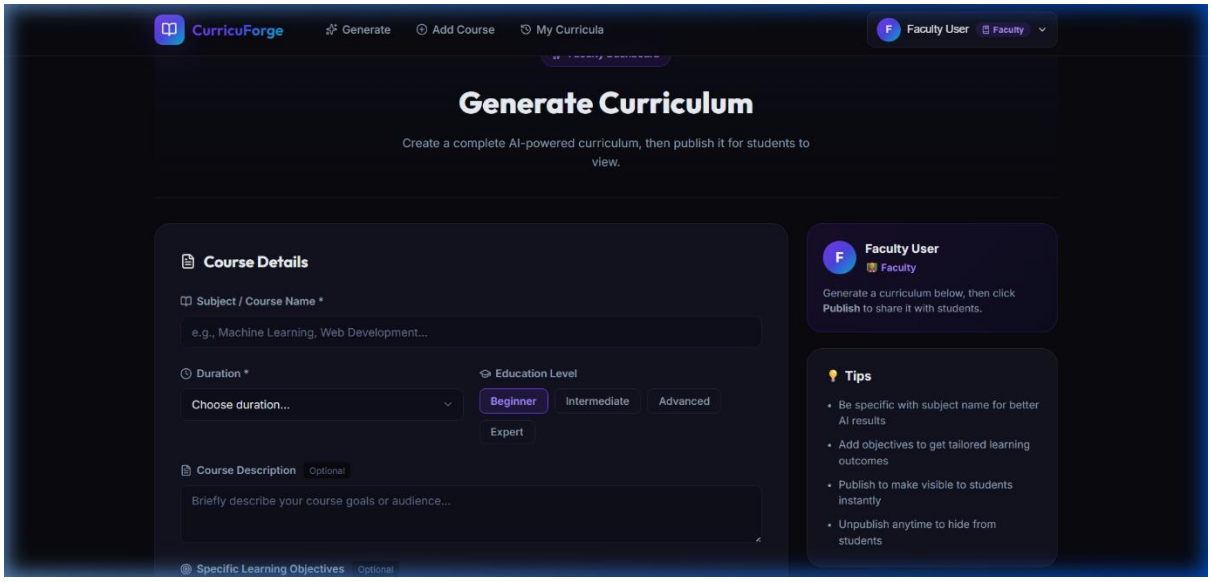
Login Page



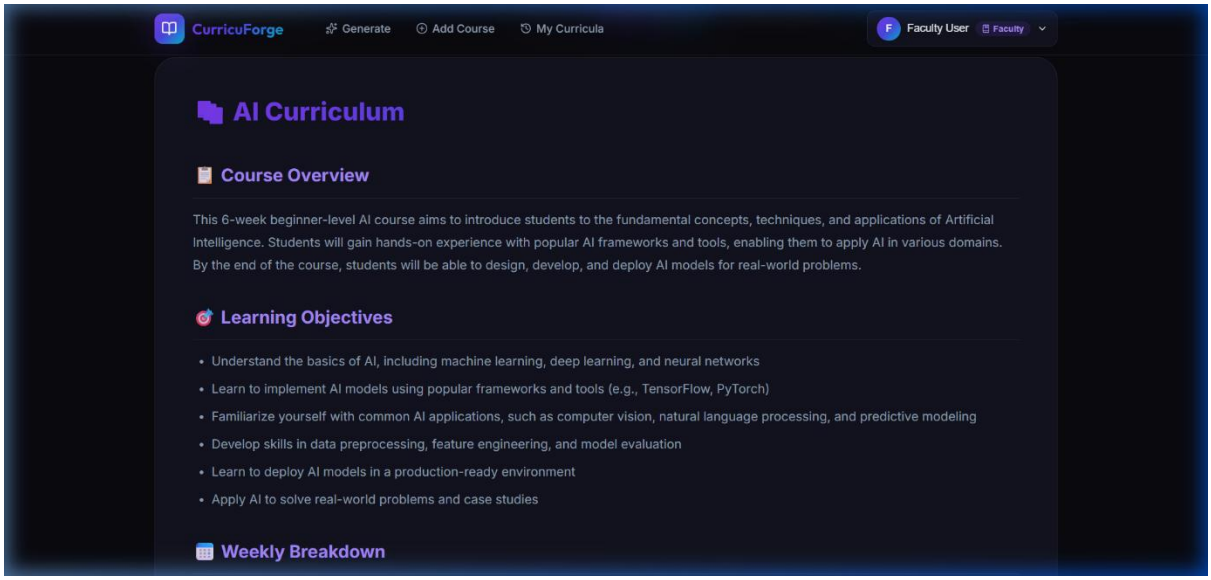
Register Page



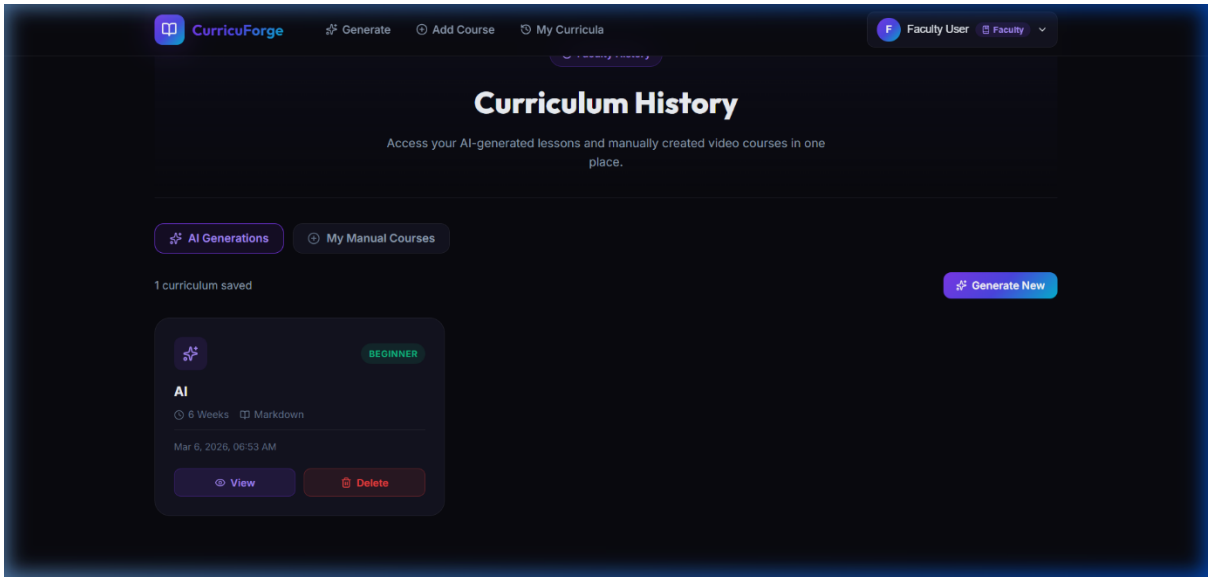
Faculty Generate Curriculum



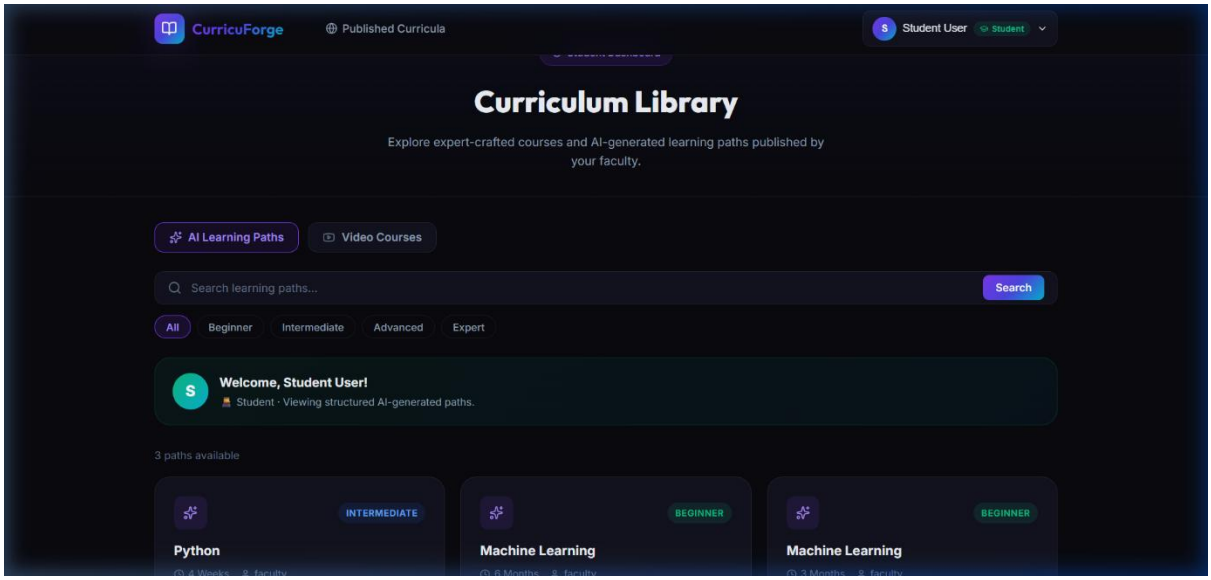
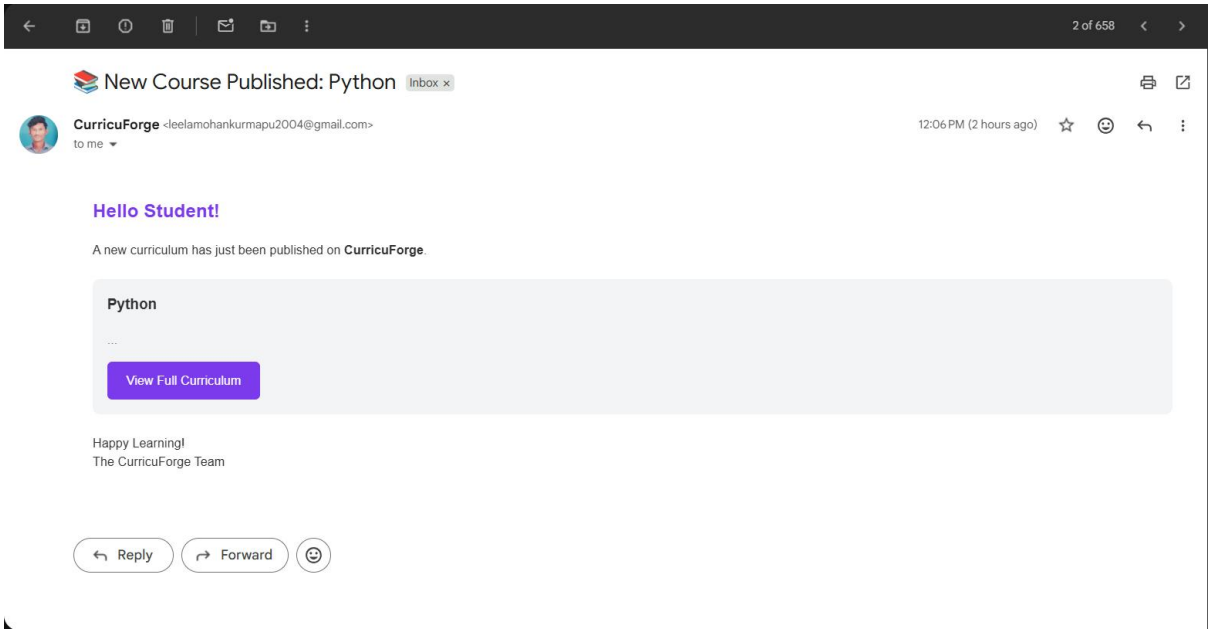
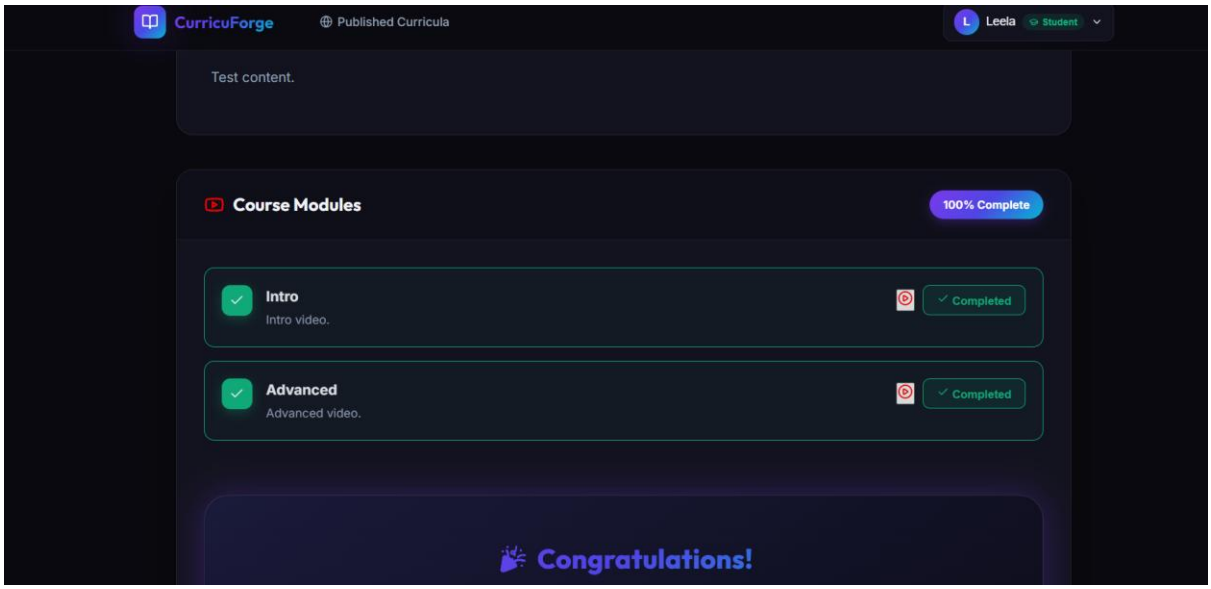
Generated Text



Curriculum History



Student Dashboard



Student Received Mail